Homework 7: 09/29 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed topic. You should be finding these in the *Exercises* section of the page.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Interpreting the Derivative

Choose 4 of the following problems to solve. Do not choose problems you've solved on previous homeworks.

1. AC 1.5.1
2. AC 1.5.7
3. AC 1.5.8
4. APEX 2.2.14
5. APEX 2.2.15-18 (each of these counts as one problem)
6. PAULS-ASSIGN.Derivatives.Interpretation of the Derivative.4-6 (each of these counts as one problem)

7. PAULS-ASSIGN.Derivatives.Interpretation of the Derivative.7-9 (each of these counts as one problem)

2 Type 2 - Estimating the Derivative

Choose 3 of the following problems to solve.

1. AC 1.5.5
2. AC 1.5.6
3. PAULS-ASSIGN.Derivatives.Interpretations of the Derivative.1-3 (each of these counts as one problem)

3 Type 3 - The Second Derivative

Choose 4 of the following problems to solve.

1. AC 1.6.1-3 (each of these counts as one problem)
2. AC 1.6.5-13 (each of these counts as one problem)
3. APEX 3.4.1
4. APEX 3.4.2
5. APEX 3.4.3 and 3.4.4 (this pair counts as one question)

4 Challenge Problem

Suppose we roll a ball whose mass is 10 grams, along a complicated track, and measure its position $p(t)$ relative to the start, in meters, as the time from when we release it t, in seconds.

We were only able to sample the distance $p(t)$ at the following points:

t (sec)	$p(t)$ (m)
0	0
1	10
2	15
3	12
4	17
5	25
6	45
7	30
8	20

Figure 1: Measurements of the Position $p(t)$ as t varies

Estimate the force on the ball $F = \text{mass} \cdot \text{acceleration}$ at time $t = 4$ seconds. Explain why you believe your approximation is the best possible.